Lecture 00 / teaching notes

Introduction to cryptography • 11 slides • Suggested delivery: 11 minutes 45 seconds, with room to adapt within a 10–15-minute opening.

Purpose and tone

Give students a reason to care before introducing notation. Retain the reference's broad, inviting survey: what cryptography protects, how its uses evolved, where it appears in daily life, and why secure systems require more than cryptography. No formulas or algorithm descriptions are needed here.

Slide	Main idea	Time	Elapsed
1	Cryptography	0:25	0:00–0:25
2	You used it before class.	0:55	0:25–1:20
3	Three questions behind digital trust.	1:30	1:20–2:50
4	New tools. New ways to communicate.	0:55	2:50–3:45
5	Private words. Public networks.	1:20	3:45–5:05
6	How do strangers trust a connection?	1:20	5:05–6:25
7	One tap. A chain of trust.	1:05	6:25–7:30
8	Your face can unlock a key.	1:10	7:30–8:40
9	Trust travels with the data.	1:05	8:40–9:45
10	Cryptography is part of the system.	1:25	9:45–11:10
11	Next: ciphers you can see inside.	0:35	11:10–11:45

Delivery

Keep the opening cover brief. On slide 2, invite one or two examples students used that morning. Spend more time on the three protection goals and the final system-level limitation. Avoid turning the application slides into protocol tutorials. The final slide leads directly into Lecture 01 on classical ciphers.

Explanations and accuracy notes

The three protection goals

Confidentiality asks who can read the content. Integrity asks whether it has been altered. Authentication asks who or what is on the other side, or where a message came from. Cryptanalysis studies how protections can fail. These are different questions; secrecy alone does not answer all of them.

History and changing technology

Connect hand writing, machine-assisted military communication, and digital networks. Enigma cryptanalysis drew on Polish work and multinational Allied teams. Avoid attributing the history to a single person. The historical thread should motivate the subject, without adding a new cipher lesson.

Messaging and the web

The messaging illustration describes an end-to-end encrypted conversation, not all chat or ordinary email. The forwarding service is not intended to have the keys needed to read the content. Metadata, backups, and compromised devices are separate concerns. HTTPS authenticates the named domain and protects a connection; it does not certify the operator’s honesty. The domain bank.example is fictional.

Payments and passkeys

Cryptographic payment checks help authenticate transaction data and detect changes. Payment methods do not all use identical mechanisms, and these checks do not prevent every fraud. With passkeys, a local face, fingerprint, or PIN check can authorize use of a credential; the website receives cryptographic proof, not the biometric as its login credential. Some passkeys sync between devices.

Passports and signed updates

A signature can support checks of data origin and integrity. An electronic passport’s chip-data signature does not by itself identify the person holding the document or prove that a chip was not cloned. A software signature is not a guarantee that the program contains no bugs. These qualifications belong in spoken explanation when relevant.

One clear limitations discussion

The transfer scam is illustrative, not a news report: a person can be manipulated into authorizing a genuine transaction whose cryptographic checks succeed. Secure devices, software, key handling, people, and processes also matter. Keep this point in one slide rather than repeating warnings after every application.

Bridge to the first technical lecture

The final glimpse MEET → PHHW uses a shift of three, but do not derive it here. Invite curiosity about how a message can be hidden and how another person might recover it. Lecture 01 supplies the construction and analysis.

Sources and updates from the reference

The source records distinguish material inspected in the uploaded reference, documentation retrieved from official repositories, and background standards cited bibliographically. The deck does not depend on live news or rapidly changing statistics.

[R] Teaching reference

Wu Hongjun, supplied Lecture01_Introduction.pdf, 12 pages.

Uploaded reference inspected. Gist retained; original PDF not redistributed.

[1] HTTPS and TLS

Mozilla / MDN, HTTPS glossary and Transport Layer Security guide.

https://developer.mozilla.org/en-US/docs/Web/Security/Defenses/Transport_Layer_Security

Supporting text retrieved from the official mdn/content GitHub repository.

[2] Encryption and messaging

Mozilla / MDN, Encryption glossary; Signal Android official project README.

https://developer.mozilla.org/en-US/docs/Glossary/Encryption

Encryption definition and official messenger project description retrieved. The slide uses a generic end-to-end encrypted conversation, not a claim about all apps.

[3] Passkeys

W3C, Web Authentication specification; Mozilla / MDN, Passkeys guide.

https://developer.mozilla.org/en-US/docs/Web/Security/Authentication/Passkeys

W3C specification source and MDN guide retrieved from their official GitHub repositories.

[4] Signed updates

The Update Framework, specification, motivation and metadata verification.

https://theupdateframework.github.io/specification/latest/

Specification source retrieved from the official theupdateframework/specification GitHub repository.

[5] Payment and passport background

EMVCo, EMV chip/contactless specifications; ICAO Doc 9303, Part 12, Security Mechanisms for MRTDs.

Background standards cited bibliographically; no usage statistics, regional rollout dates, or claim of a newly published version.

What was refreshed

Retained the three security goals, the connection to computing and communication, military history, finance, daily-life applications, and the system-level limitation. Replaced the Swift-volume and regional ATM discussion with timeless payment concepts; refreshed account authentication with passkeys; added signed software updates; and used a clearly illustrative scam instead of relying on the 2016 Bangladesh Bank incident. The original uploaded PDF is not redistributed.